

NextGen Exoskeleton Navigator Technical Report
NextGen Exoskeleton Navigator Risk Management Exercise
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Executive Summary

The NextGen Navigator (NGEN) system is a state-of-the-art exoskeleton system with capabilities of manned, unmanned, and autonomous modes with the means to support security, research, and engineering personnel. As the pinnacle of NextGen technology, the importance of identifying and mitigating technical risks cannot be understated in their deployment groups.

Risk assessment and risk management are critical functions of proper cybersecurity practices in ensuring successful business operations. The assessment, categorization, and documentation of risks will prevent financial losses, optimize security spending budgets, and ensure compliance with legal and regulatory bodies presiding over NextGen. Documented within this report are risks deemed important and relevant to the success of NextGen while conforming to an exceptional cost-effectiveness. Provided with these risks are the overall likelihood of risk realization allowing for enhanced decision-making and proper planning to be undertaken.

In conclusion, the NextGen Exoskeleton Navigator System is the culmination of the greatest minds of our time to envision a product capable of propelling NextGen even further ahead of industry competition all while revolutionizing research, autonomous drone use, and security implementation. This report provides a detailed breakdown of fundamental and critical technical risks that the NextGen Navigator Exoskeleton and their deployments face while enabling stakeholders to understand the threats the deployments face in the field.

Discussion

The risks identified in this risk assessment were determined through a meticulous examination of the NextGen Navigator system by a determined and tired risk analyst well-versed in the system's deployment architecture and the project's software requirements. The assessment was framed with the National Institute of Science and Technology (NIST) cybersecurity framework in mind, which alongside the OWASP Risk Rating Methodology for analyzing business impacts, provided a structured approach for evaluating the risks associated with NGEN deployments when operating within field environments.

By utilizing these frameworks, a comprehensive analysis of the NGEN system was conducted to evaluate the potential impacts on business functions, reputation, privacy policies. Furthermore, the risk assessment considered the level of non-compliance that could result from each identified risk. This thorough approach ensured a clear and accurate representation of the risks facing the NextGen Navigator system, enabling stakeholders to make informed decisions regarding risk prioritization and risk mitigation strategies to ensure the confidentiality, availability, and integrity of NextGen data and systems.

Technical Risks

Technical Risk I

Technical Risk 01 was identified as a critical risk to business functions given the significant threat it poses to the system's overall functionality and security. With direct impacts to confidentiality and availability, TeR 01 is a risk that should be further evaluated with mitigation techniques employed to prevent this risks realization. Points of realization revolve around user-to-machine interfaces and the session management software contained within the NGENius logic layer.

Technical Risk 1

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 01	Failure to properly transition between pilots sessions and their designated permissions.	Potential financial costs due to tool misuse and system / property damage. Possible non-compliance violations in the event security personnel tools are accessed.	Remote or physical session management interfaces within NGEN Navigator Exoskeleton suits.

InfoSec Triad	Description
Confidentiality:	Unauthorized access to pilot session data which contains private personal identifying information. Impacts the confidentiality of pilot data and personal identifying information.
Availability:	Unauthorized use of tools by personnel inexperienced with their functionality may damage the NGEN Exoskeleton tools and NextGen property impacting the availability of role specific equipment.
Integrity:	Non-applicable

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
3	9	5	0
Overall likelihood = 4.25			

Technical Risk II

Technical Risk 02 was identified as a fundamental risk in the NGEN Exoskeleton deployment scheme given its wide-reaching impact should the risk be realized. NGEN Exoskeleton communications to their designated Drone Operation Centers contain personal information as well as sensitive business information that would have lasting impacts on business functions as well as compromising the confidentiality of NextGen data.

Technical Risk 2

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 02	Threat actor interception of communications on insecure lines between NGEN Exoskeletons and their designated DOCs.	Possible financial costs due to reputational damages as well as non-compliance charges.	Wireless or physical data transmission links between NGEN Navigator Exoskeletons and Drone Operations Centers (DOC)

InfoSec Triad	Description
Confidentiality:	Insecure communication links between NGEN Exoskeletons and DOCs has the potential to impact the confidentiality of business information via threat actor eavesdropping, recording, and leakage of personal identifying information, confidential data, and / or restricted data.
Availability:	Non-applicable
Integrity:	Non-applicable

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
7	9	7	9
Overall likelihood = 8			

Technical Risk III

Technical Risk 03 was identified as a critical risk to business functions due to the value of NGEN Exoskeletons, their capabilities, and their direct communications to DOC and in turn Command Centers. The realization of TeR 03 would have a direct impact on the confidentiality, availability, and integrity of NextGen data by means of access to local database storage onboard NGEN Exoskeleton systems, as well as the potential destruction of suit attachments leaving systems in a damaged state.

Technical Risk 3

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 03	Unauthorized access to NGEN Exoskeletons by non-NextGen personnel.	Possible unauthorized use of NGEN property, tools, and resources resulting in financial costs due to asset and reputational damage.	Remote or physical session management interface within NGEN Navigator Exoskeleton suits.

InfoSec Triad	Description
Confidentiality:	Without secure authentication, unauthorized users may gain access to sensitive information, compromising the confidentiality of NextGen critical data.
Availability:	If unauthorized users gain access, they may misuse the system or cause disruptions, impacting the availability of the system for legitimate users
Integrity:	Unauthorized access to NGEN exoskeletons provide direct access to local data storage potentially impacting the integrity of data on local systems.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
3	5	7	3
Overall likelihood = 4.5			

Technical Risk IV

The risk of a power system failure in a NGEN Exoskeleton is a risk that upon its realization, could cripple critical business functions on NGEN deployments in remote areas. A loss of power or brownout in a NGEN Exoskeleton would directly impact the availability of the systems accessibility due to lack of operation controls and could affect the integrity of locally stored data that has yet to be transmitted to its DOC by means of data corruption or destruction.

Technical Risk 4

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 04	NGEN Exoskeleton system power failure	Potential financial costs due to asset damage or destruction and personnel harm which would lead to reputational damage. Loss of productivity due to hinderance of business processes.	NGEN Navigator Exoskeleton suit power supply / system.

InfoSec Triad	Description
Confidentiality:	Non-applicable
Availability:	Power failure in a NGEN Exoskeleton system would directly impact availability in that the system would be unresponsive and local session data would either be destroyed or inaccessible.
Integrity:	Power failure or brownout incidents with NGEN Exoskeleton systems has the potential to corrupt session data as well as local database storage units, impacting the integrity of NextGen data contained with units.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
2	1	2	0
Overall likelihood = 1.25			

Technical Risk V

Technical Risk 05 has been identified as a critical risk to business functionality due to the criticality of NextGen data housed within the research, engineering, security, access control, and NGEN configuration databases. The use of depreciated encryption standards provides threat actors with an additional attack surface, which should this risk be realized would inherently impact the confidentiality of data-at-rest and the integrity of the data due to an inability to ensure that the encryption had not been broken and data tampered with. Impacts of this realization range from financial costs to extreme privacy and non-compliance charges, as well as reputational damages due to negligence.

Technical Risk 5

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 05	Data breach as a result of the use of depreciated encryption standards.	Potential financial, privacy, non-compliance, and reputational damages should a breach of NGEN Navigator, DOC, or Command Center databases occur.	Databases locally housed within NGEN Navigator Exoskeletons, Drone Operations Centers, and / or Command Centers.

InfoSec Triad	Description
Confidentiality:	Data breaches will incur the dissemination of NGEN private data of different class types impacting the confidentiality of personnel, systems, and other business function data.
Availability:	Non-applicable
Integrity:	The use of poor encryption standards may directly or indirectly affect the integrity of NextGen data at rest by means of tampering due to its poor security while at rest within NextGen databases.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
6	9	7	9
Overall likelihood = 7.75			

Technical Risk VI

Technical Risk 06 was identified as a fundamental risk that could be realized within NextGen systems that has an extreme priority. NextGen as an organization operates within the confines of multiple nations and has information classification levels ranging from internal to restricted that must be protected and kept confidential. Should an actor's actions be unaccounted for, threatening or harmless, the integrity and confidentiality of NextGen data cannot be guaranteed.

Technical Risk 6

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 06	Unsupervised and untraceable activity on NextGen software systems.	Potential financial and privacy charges and damages.	User access surfaces which interface with NextGen Navigator systems such as workstations and exoskeleton GUIs.

InfoSec Triad	Description
Confidentiality:	Inadequate logging, traceability, and accountability provides poor means of tracing user actions and holding them accountable for changes made, or events created within NextGen systems resulting in the confidentiality of data on devices being potentially impacted.
Availability:	Non-applicable
Integrity:	Poor traceability of actions taken on NextGen Navigators systems could impact the integrity of confidential and restricted data due to an inability to determine who the accessor was, what they did to the data, and where they interacted with the data.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
2	4	5	9
Overall likelihood = 5			

Technical Risk VII

Technical Risk 07 has been identified as a critical risk to business functionality due to its prolific history in the world of software engineering. Systems that do not fail gracefully and reveal internal info-system information have the potential to create additional threat vectors for malicious users attempting to penetrate NextGen systems. Should this risk be realized and taken advantage of unauthorized entry into the network may be possible as well as the breach of the confidentiality, integrity, and availability of systems compromised by attackers. Point of realization for TeR 07 are user access points into the NextGen network such as workstations, Exoskeleton login GUIs provided by NGENius, et cetera.

Technical Risk 7

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 07	Unauthorized threat actor access via inadequate handling of software errors.	Has the potential to lead to financial and reputational damages through privacy violations and the exfiltration of data.	Software system interfaces such as NGENius login screens and NGEN Exoskeleton interfaces.

InfoSec Triad	Description
Confidentiality:	Failure to fail gracefully has the potential to impact the confidentiality of internal data by means of providing threat actors information on internal system configurations and versions enabling the penetration of NextGen software.
Availability:	Availability could be impacted by means of not handling failure properly resulting in the crashing and loss of availability of a NextGen software system.
Integrity:	Failure to fail gracefully can result in the corruption of session data and data in use at the corresponding interface, directly impacting the integrity of the data.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
3	4	2	7
Overall likelihood = 4			

Technical Risk VIII

Technical Risk 08 has been selected as a high priority risk due to its simplicity and cost-effectiveness in relation to its extreme business impacts and impacts to data confidentiality, availability, and integrity. The realization of a threat actors introducing malicious hardware via system ports could spell disaster for NextGen data and systems. This risks realization would be a pathway to other attacks of immense scope.

Technical Risk 8

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR 08	Threat actors inserting malicious USB drives or other hardware to deliver a payload within NextGen software systems.	Risk realization could result in the loss, corruption, and destruction of critical business data, the breaching of personnel and / or business data, and the destruction of NextGen property.	Human to software interfaces such as NextGen workstations, NGEN Exoskeletons, personnel mobile devices, and printers.

InfoSec Triad	Description
Confidentiality:	Vulnerable system ports have the potential to directly impact the confidentiality of data dependent on the payload delivered within NextGen systems via its realization. This threat vector could provide a means of data exfiltration where its encryption can be broken and its confidentiality impacted.
Availability:	The availability of systems affected by the realization of this risk may be impacted by means of a denial-of-service attack, ransomware, or malicious software crashing the information system entirely.
Integrity:	The integrity of systems affected may be directly impacted dependent on the payload the malicious actors inserted into the system via the vulnerable physical port. Ransomware, viruses, and other malicious software can directly affect data at rest changing their contents or corrupting it entirely.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
7	4	5	7
Overall likelihood = 5.75			

Technical Risk IX

Technical Risk 09 has been selected as a fundamental risk to NextGen systems and its business functions due to the proliferation of internal threats throughout the business world. Unruly and disgruntled employees make up 62% of data breaches in 2024 and should be recognized as a critical risk. Indirect impacts to confidentiality, availability, and integrity are a possibility should this risk be realized due to the nature of their employment within the company and the access they may or may not have access to.

Technical Risk 9

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR-09	Internal threats leading to privilege abuse, sensitive data breach, and asset destruction.	Risk realization could result in the exfiltration of critical business data resulting in financial and reputational damages.	Internal user to software interfaces such as NextGen workstations, servers, and terminals.

InfoSec Triad	Description
Confidentiality:	Realization of this risk could result in the confidentiality of NextGen data being impacted by means of internal employee sabotage, data breach, or asset and data destruction.
Availability:	The availability of systems affected by the realization of an internal threat actor could be directly impacted by means of asset destruction or tampering.
Integrity:	Internal threat actors could directly impact the integrity of data dependent on the privileges acquired by the actor in question.

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
7	9	5	9
Overall likelihood = 7.5			

Technical Risk X

Technical Risk 10 has been identified as a critical risk due to the nature of NGEN deployments and the operating environments encountered by them. A robust and stable means of communication must be ensured while operating NGEN Exoskeletons in the field whether it be remotely or physically. Impacts to availability would be directly impacted should this risk be realized while on deployment with pilots and remote operators unable to properly control drone behavior and movements.

Technical Risk 10

Risk ID	Technical Security Risk	Business Impact(s)	Risk Realization Point
TeR-10	Failure of network communication links due to natural disasters such as earthquakes, hurricanes, et cetera, resulting in the loss of communication between NGEN Exoskeletons and their designated DOCs in the field.	Financial damages due to loss of productivity, asset damage or destruction in the form of NGEN Exoskeletons and NextGen personnel and inability to remotely monitor and control Exoskeletons.	All NGEN deployment sites operating within the confines of Earth.

InfoSec Triad	Description
Confidentiality:	Non-applicable
Availability:	The loss of communication links between DOC and NGEN Exoskeletons would directly impact the availability of Exoskeletons, DOCs, their data, and their communications to other NextGen facilities.
Integrity:	Non-applicable

Business Impact			
Financial Damages	Reputational Damages	Non-compliance	Privacy Violation
5	0	0	0
Overall likelihood = 1.25			

Conclusions and Recommendations

The NextGen Exoskeleton Navigator (NGEN) system is a revolutionary technology that will change the way we think about robotics, autonomous systems, and biorobotics. With deployment environments of the best conditions to the worst conditions, the foundations to inception of this system are critical for its success. The risks identified, organized, and presented are a curated selection that threaten the stability of the NextGen Navigator deployments. The overall success of this project is of paramount importance, and such every risk should be examined with the utmost scrutiny. The impact on the secure software development of this project, while quite substantial, ensures that security be implemented at all stages of the software development lifecycle (SDLC). Security should not be an afterthought and should be implemented from the top down and bottom up. With this in mind, all areas of the project were examined with a wide range of risks derived to allow for more coverage, but all areas should consistently be revisited and reevaluated to ensure that as risks are mitigated emerging risks are identified, documented, and categorized to continue an ongoing emphasis on security.

Of the risks assessed within this report there is a top-level risk that must be dealt with, and one lower on the spectrum that regardless of its ranking, should still be treated as a genuine risk to NGEN deployments:

Highest Priority Risk:

- Technical Risk 02: Threat Actor exfiltration of cleartext communications between NGEN Exoskeletons and DOCs.

Lowest Priority Risk:

- Technical Risk 10: Communication failures due to natural disasters and incursions.

Recommendations:

- **Continuous Risk Analysis:** The concept of security is not something to be done once and forgotten about. Continuous analysis of hardware, software, business processes, system architecture and their risks should be undertaken throughout the development process from start to finish, as well as after their production launch. This will ensure a solid foundation for the business to properly function and continue to be profitable.
- **Mitigation Analysis and Implementation:** With risk assessment comes mitigating them. This document was an assessment of risks, leaving the mitigation techniques to be employed to a future report to allow for more NextGen exercises.
- **Incident Response Planning:** Incident response is a critical functionality of business systems. With this, incident response plans should be routinely updated and developed as needed (Adsero Security).
- **Disaster Recovery Planning:** Disasters are always a possibility, and the risk rating of threats will always be non-zero. From this, disaster recovery plans should be developed and consistently revisited to ensure the success and recovery of the NGEN project should disaster strike (Adsero Security).

References

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